

Rehab? Enter Wiihab!

Using gaming technology in healing. Read how game consoles, particularly the Wii, is being used as a therapeutic tool

**Nintendo's new Gamepad**

Nintendo unveils a new tablet computer sporting a 6.2 inch touchscreen Wii U gamepad <http://dgit.in/L1qnxz>

God is a programmer

We might be living in the most elegantly coded virtual reality ever - the Universe itself!

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There comes a time in a man's life when he feels that his life is building up to something. Everything around him seems to be plotting a path that his life will take. When this realisation dawns, he feels a strange unrest about him - a feeling that someone, somewhere is observing him, that somebody has told the Universe his plan for the man and there is nothing he can do about it. We all have our "Truman moments", sometime or the other. In Douglas Adams' historic masterpiece, *The Hitchhikers' Guide to The Galaxy*, Arthur Dent always felt an 'unaccountable feeling in his bones that something sinister was going on in the Universe, but no one would tell him about it'. That was also the feeling Keanu Reeves portrayed brilliantly in the movie franchise *The Matrix*. As it turns out, it might well have been a documentary - someone probably has told the Universe his plan for each one of us, and it's not God that we're talking about here. It might be a programmer of a higher level. You read that right.

How this came to be discovered

Recently, a scientist at the University of Maryland claimed to have found computer codes embedded into the equations of string theory. Now, string theory is what researchers have their hopes pinned on, to be proved as the elusive "Theory of Everything", and indeed it has the most valid claim for being exactly that. The problem was that until recently, it had been a circular logic - a set of equations and assumptions, where everything

bases its own existence on the other, and the entire system exists as a mutually dependent framework. The drawback was that this framework could never be proven to be, in any way, related to our own, real world. That it describes many phenomena very accurately is mostly because it was designed to do exactly that - be the theory of everything, in *this* world. In fact, this has been one of the most persistent criticisms of the theory - that no situation could ever be imagined that would ever prove (or even disprove) the idea. That was until physicists at Imperial College London had a flash of imagina-



Dr. Sylvester James Gates

Dr. Sylvester James Gates is an American theoretical physicist whose research fields include Syperymmetry, Supergravity, and Superstring Theory. Along with being a senior professor at the University of Maryland, he serves on the America President's Council for Advisors on Science and Technology

tion, and basically proved the existence of string theory in the universe, albeit in a context very different to the one it was being applied to. They saw that the kind of mathematical application that string theory puts forward for the explanation of black holes is very similar to the one that was being used to explain certain quantum systems, called Qubits. Thus, when applied to a field that is more related to high speed data transfer than being the explanation to everything, string theory came out with flying colours. By the alleged symmetry in nature, it was only logical that this gave a ray of hope to physicists that string theory may actually

have a proof after all. Research work in this field received unprecedented impetus. One of the researchers, Dr. Sylvester Gates, approached the task at hand with a different methodology. By his own admission, he is not a big fan of using long, scary equations in physics. Instead, he and his fellow researchers made use of graphical representations called 'Adinkras', which essentially put out a pictorial relation between the elements of the equation. And when this method was applied to the equations of string theory, something strange was observed. The equations seemed to have codes built into them. Not just any code, specific computer codes. And not just any normal computer code - they contained a special kind of code, known as the "Doubly-even self-dual linear binary error-correcting block code" (these are not as far removed from us as they sound - they are built into all of our modern Internet browsers). These are essentially codes that make sure that data doesn't undergo corruption during its transmission from point A to B. And it's now known that they are built into the very fabric of our universe too (that is, assuming string theory correctly describes the fabric of our universe). Read more in the essay 'Symbols of Power' at <http://dgit.in/symbolsofpower>. Read Dr. Gates' research paper at <http://dgit.in/drgatespaper>.

ANTHROPIC PRINCIPLE

Anthropic Principle states that a universe must be compatible with the conscious life form that observes it. That means, we observe the universe this way, because this was how it needed to be in order for us to be there and observe it. It is like saying that everything has been finetuned for our existence. One of the most compelling of arguments by theists about the existence of God; it can just as easily support the possibility of there being a 'mythological' programmer instead. A common retort is that just as a monkey could type a Shakespearean sonnet if given enough time, similarly the universe would've stumbled upon one or the other life sustaining condition, given sufficient entropy.